

Low-Power VLSI Design for Edge AI Accelerators in IoT Devices

Nivitha G
Student

Department of Electronics and
Communication Engineering
Saveetha Engineering College
Chennai, Tamil Nadu, India
nivithag.1007@gmail.com

Aravind T
Professor

Department of Electronics and
Communication Engineering
Saveetha Engineering College
Chennai, Tamil Nadu, India
aravindthangavelme@gmail.com

R. Saravanakumar
Software Analyst

Iconix Software Solution
Tirunelveli, Tamil Nadu, India
iconixsaro@gmail.com
ORCID: 0009-0003-2343-2308

Abstract—The paper's main goal is to provide an energy-efficient hardware architecture that can run machine learning activities directly on edge devices with little delay and power expenditure. As the need for smart IoT apps develops, Artificial Intelligence (AI) accelerators that work well with Very Large Scale Integration (VLSI) are needed for real-time data processing with stringent power constraints. Traditional AI computation on cloud platforms has security concerns and latency since data is always being sent. So, to move AI processing to the edge, we need small, low-power VLSI architectures that can conduct neural calculations quickly. Conventional accelerators have a lot of switching activity, waste a lot of power, and can't be easily scaled up on Field-Programmable Gate Array (FPGA) and Application-Specific Integrated Circuit (ASIC) platforms; thus, they aren't good for IoT edge situations. The goal of this project is to use VHDL on Xilinx ISE 14.7 to design and build a low-power AI accelerator that improves area, time, and power while keeping computational accuracy. The design uses clock gating, reducing the width of operands, and sharing resources in a matrix multiply-accumulate (MAC) architecture. A control Finite State Machine (FSM) and gated clock logic keep the operation flow running smoothly, while ISim is used to test functionality via simulations. The suggested architecture lowered dynamic power use and area use while keeping throughput the same, showing that it is good for real-time edge AI activities on Internet of Things (IoT) devices.

Keywords— *Energy-Efficient, Machine Learning, Edge Devices, Very Large Scale Integration, Data Processing, Multiply-Accumulate, Finite State Machine, Application-Specific Integrated Circuit, Field-Programmable Gate Array, Internet Of Things, Artificial Intelligence*

I. INTRODUCTION

Edge AI robotics increasingly relies on deep neural networks that must operate in real time on the device despite adhering to stringent power and silicon constraints, without compromising the speed or accuracy of jobs. Despite the continuous advancements in VLSI, system-on-chip, and sector configurable gate array design, numerous embedded accelerators continue to squander energy due to the high cost of memory transfer during inference, underutilization of processing elements when workloads transition from dense to sparse, and the lack of end-to-end connectivity for verification and measurement interfaces. In reality, this indicates that system-level power may remain elevated despite satisfactory nominal compute consumption, and latency targets may be unmet due to memory stalls rather than limitations on arithmetic throughput. In robotic environments, these issues are exacerbated because of

fluctuating input scales, frame rates, and modeling selections throughout operation, coupled with limited thermal capacity imposed by compact designs.

The growth of the IoT has made it far more important for algorithms that analyze digital data and employ machine learning to run on edge devices that are both fast and energy-efficient. The VLSI design of edge AI processors for IoT devices is a new way to solve this challenge. Investigating the VLSI architecture of these processors is crucial as it facilitates seamless deployment in remote locations and provides energy-efficient solutions for integrated AI applications [1]. This contemporary, multidisciplinary area includes hardware design, chip architecture, embedded technology, machine learning, digital signal processing, low-power circuit layout, power control, and VLSI design of edge AI processors. The goal of the studies in this field right now is to create efficient architectures that use less space and energy than standard processors [2]. These ideas combine fast binary mapping techniques with multiport storage, registry storage, and mathematical platforms. Also, they utilize fine-grain parallelism to get maximum flavor out of the grain while using the least amount of juice. Researchers are having a hard time designing low-power edge AI systems that use VLSI [3].

A. Background and Motivation

The IoT is expanding rapidly, with billions of interconnected devices already gathering and processing vast amounts of data instantaneously. AI computations conducted on cloud servers encounter latency, bandwidth constraints, and privacy vulnerabilities. This renders cloud-only solutions ineffective for applications requiring rapid execution. Edge AI has emerged as a remedy to these issues. It enables IoT devices to process data intelligently at the device level. To implement complex algorithms for AI at the edge, one needs electronics that are energy-efficient and high-performing, capable of operating under the constraints of space and power. This necessitates improved VLSI architectures that minimize dynamic power consumption while maintaining high computational speed. Employing VHDL on FPGA devices facilitates the prototyping and testing of such systems with ease. Subsequently, ASIC implementations may exhibit greater efficiency for extensive applications [4]. The objective of this project is to develop a low-power AI accelerator that operates efficiently via the implementation of FSM control, clock restriction, and allocation of resources. This architecture facilitates the development of durable Edge AI solutions applicable across many next-generation IoT applications, such

as smart healthcare, self-driving devices, and environmental monitoring, by enabling real-time inference with reduced energy consumption.

B. Contribution

- This paper presents a power-efficient VLSI accelerator architecture tailored for edge AI applications in IoT devices, overcoming the shortcomings of traditional accelerators that often exhibit elevated power consumption and lack optimization for resource-constrained settings.
- The suggested design has enhanced multiplier-accumulator (MAC) units and efficient dataflow mechanisms, which markedly decrease switching activity and computational overhead, thereby enhancing energy efficiency and overall processing speed.
- The suggested architecture, in contrast to conventional AI accelerators intended for cloud or high-performance systems, emphasizes a lightweight hardware implementation characterized by diminished silicon area, decreased latency, and minimal power consumption, making it appropriate for real-time IoT edge devices.
- The research includes a comparative benchmark analysis using established hardware performance indicators, including power consumption, latency, throughput, and energy per operation. The findings indicate that the proposed architecture significantly outperforms current methodologies.
- The suggested architecture is confirmed by simulation and hardware assessment, confirming its viability for next-generation low-power edge AI accelerators in IoT systems.

II. LITERATURE REVIEW

A thorough examination of neuromorphic computing models and hardware architectures, with a focus on computation inspired by the brain for AI systems that are both energy efficient and driven by events. Their research discussed the possibility of mixed-signal VLSI implementations, memristor-based circuits, and spiking neural networks (SNNs) for AI edge applications that are both power-efficient and run in real-time. Reviewing state-of-the-art designs like SpiNNaker, Loihi, and TrueNorth was the key technique. On the other hand, neuromorphic systems have a hard time fitting into traditional Internet of Things (IoT) ecosystems due to their lack of scalability and standardization [5].

An analysis was conducted on digital and analog AI accelerators, comparing and contrasting their energy efficiency, scalability, and accuracy. Digital CMOS-based designs and analog in-memory computing structures using memristors were both architecturally evaluated as part of the process. Trends in AI inference architecture, such as hybrid digital-analog accelerators, were brought to light. The main problem with digital accelerators is their reduced density and increased energy consumption, which prevents them from being used in low-power Internet of Things devices. Analog accelerators, on the other hand, are sensitive to noise and manufacturing variability [6].

A small FP16 matrix multiplication accelerator for adaptive deep learning on RISC-V systems on chips (SoCs) called ReDMuLE is presented. To reduce computing power usage, their solution made use of adaptive precision management and hardware-efficient MAC arrays. The accelerator improved accuracy without sacrificing throughput when deployed on an extremely low-power platform. On the other hand, it has limited performance scalability for big neural networks and relies on FP16 arithmetic, which can lower accuracy for more sensitive AI models [7].

An effective method for convolutional dataflow in low-power neural network accelerators called EcoFlow was presented. To reduce memory access and maximize compute efficiency in convolutional neural network (CNN) inference tasks, the approach centered on data reusing and optimizing pipelines. It showed that different hardware accelerators significantly reduced power consumption. The effectiveness of EcoFlow is very workload and neural network model-dependent, which limits its adaptability to various Internet of Things (IoT) situations and may need architectural redesign [8].

An approach for adaptive design that minimises power loss in nanoscale very large scale integration (VLSI) systems by accounting for fluctuations in process, voltage, and temperature (PVT). In order to achieve a balance between performance and leakage, the technique used transistor size and multi-threshold voltage optimization. The downside is that it made designs more complicated and required a lot of simulation and calibration across PVT corners, which isn't ideal for systems that need to be reconfigured quickly, such as FPGA-based AI accelerators, or for systems that operate in real-time [9].

Digital logic implementation using a behavioral SPICE model of a binarized memristor. For digital computing, the approach validated the switching properties of logic gates based on memristors via simulation. This method showed promise for designing ultra-low-power logic. The short lifespan and unpredictability of memristor devices are the main issues that prevent their widespread use in large-scale VLSI applications and their dependability in a variety of environments [10].

With an emphasis on hardware-software co-design, near-memory computing, and quantized neural processing, it delved into new technology and design issues for low-power AI IoT systems. They looked at several AI frameworks that are incorporated into IoT designs and how they use low-power resources. One major problem is that these systems need a lot of effort to keep the AI models in sync with the hardware limitations, and it's not always easy to maintain accuracy while saving energy [11].

Putting TensorFlow Lite machine learning models into practice on ultra-low-power microcontrollers. As part of their process, they quantized and pruned models before deploying them on embedded systems like ARM Cortex-M and Arduino. This method made it possible to do energy-efficient inference at the edge in real time. The biggest problem is that TinyML models can't handle dynamic workloads or complicated deep learning tasks because of their memory and computational limitations [12].

This is an acoustic feature extractor for always-on keyword detection devices that is tolerant to background noise and process fluctuation and uses spike-domain divisive-

energy normalization. Adaptive spike encoding and neuromorphic signal processing were the key components of the approach that allowed for absolutely lightweight operation. Even though it works well for noise robustness, the spike-based technique isn't compatible with regular digital AI accelerators used in edge IoT devices since it needs specific hardware and training methodologies [13].

Several studies have focused on improving the voltage gain of DC–DC converters used in solar photovoltaic (PV) systems. Conventional boost converters are widely used in microgrid applications but often suffer from limitations such as high switching losses, low voltage gain, and high duty cycle operation. To overcome these issues, researchers have proposed modified topologies using switched inductors, switched capacitors, and voltage multiplier techniques to enhance the voltage boosting capability and efficiency. In this context, the study “A High-Gain DC-DC Boost Converter for Solar Microgrid Systems Using a Switching Inductor and a Capacitor” introduces an improved converter topology that increases voltage gain while reducing current ripple and switching stress. This design helps improve power conversion efficiency and system reliability, making it suitable for renewable energy-based solar microgrid systems [14].

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) have enabled intelligent mental health monitoring systems capable of analyzing user behavior and emotional patterns. Several studies have proposed AI-based platforms that analyze text messages, social media posts, and conversational data to detect signs of stress, anxiety, and depression. Machine learning algorithms combined with NLP techniques can identify emotional sentiment and behavioral changes to support early diagnosis and personalized mental health care. The study “AI Personalized Mental Health Monitoring System using Machine Learning and Natural Language Processing” proposes a framework that integrates ML models with NLP-based sentiment analysis to continuously monitor users’ mental well-being and provide personalized recommendations. Such systems enhance early detection, accessibility, and personalized mental health support through intelligent digital monitoring platforms [15].

Prior research on AI accelerators mostly emphasizes high-performance computing settings or cloud-based AI processing, where limitations related to power and hardware resources are less significant. Numerous current VLSI accelerator designs prioritize computational speed, often leading to elevated power consumption, expanded silicon area, and heightened energy per operation, hence constraining their applicability for resource-limited IoT edge devices. The review of these prior studies reveals various unsolved research difficulties.

- Many existing AI accelerator architectures are not optimized for ultra-low-power operation required in IoT edge devices.

- Prior works often focus on throughput improvement but do not adequately address the energy per operation, which directly impacts battery life in embedded systems.
- Several accelerator implementations require large silicon area and complex interconnect structures, reducing their practicality for compact IoT hardware platforms.
- Existing designs sometimes introduce higher latency due to inefficient dataflow and memory access patterns, which limit their suitability for real-time edge AI applications.
- Many previous studies do not provide standardized comparative benchmarking using widely accepted hardware performance metrics, making it difficult to evaluate their practical effectiveness.

III. METHODS AND TECHNIQUES

The goal of the suggested technique is to employ VHDL on Xilinx FPGA platforms to create an AI hardware accelerator that is both optimized and energy-efficient. In order to retain high computational performance appropriate for real-time edge applications based on AI, the design seeks to minimize both static and dynamic power consumption.

The architecture intended to enable effective real-time AI processing with low power consumption is shown in Figure 1. The MAC Unit, Shared MAC Array, Control FSM, and Clock-Gating Unit are the four main functional modules that make up the architecture. These modules are integrated to maximize data flow and energy management. The primary computing block in charge of carrying out mathematical operations essential to AI tasks like matrix multiplication and convolution is the MAC (Multiply–Accumulate) Unit. The MAC unit processes input data (In₁) first, interacting with a Register File (Regfile) that serves as a temporary storage space for operands and partial results. The Shared MAC Array subsequently receives the processed data and uses parallelized calculations to speed up deep learning inference. By dynamically regulating the clock signal sent to the MAC array, the Clock-Gating Unit successfully lowers dynamic power consumption by decreasing switching behavior when certain components are not in use. By controlling timing, data flow, and coordination between processing stages, the Control FSM (Finite State Machine) coordinates the actions of every component. When all calculations are finished, the output is finally produced at the "Out" terminal. For Edge AI IoT applications like autonomous sensors and embedded statistical systems, its modular and hierarchical architecture guarantees flexibility, low latency, and significant computing throughput while preserving an ultra-low-power footprint (Figure 1).

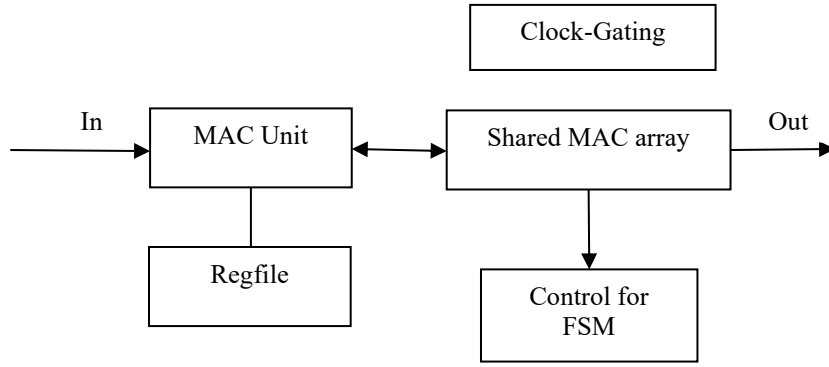


Fig. 1. System Architecture

A. System Specification and Architecture Design

The suggested Low-Power VLSI Edge AI Accelerator is made to effectively perform neural calculations in Internet of Things devices with little power. A MAC Unit, Shared MAC Array, Register File (Regfile), Clock-Gating Module, and Control FSM are all part of the modular architecture defined by the system specification. The Shared MAC Array facilitates parallel processing to speed up matrix and convolution calculations, while the MAC Unit handles the fundamental multiply-accumulate processes. In order to provide seamless data transfer between processing steps, the Regfile offers temporary data storage. In order to reduce power waste, the clock-gating module dynamically shuts off inactive circuits. Data flow control, operation sequencing, and computation stage synchronization are all governed by the Control FSM. Low space usage, fast speed, and energy efficiency are given top priority in this VHDL-designed and Xilinx FPGA-implemented architecture. This standard guarantees that the accelerator satisfies the real-time processing requirements of AI-based Internet of Things applications, including automation and smart monitoring.

B. Low-Power Design Techniques

The Edge AI Accelerator's Low-Power VLSI Design approach uses a number of cutting-edge strategies to reduce energy use while preserving great computing performance. The main method is clock gating, which reduces needless switching activity and dynamic power waste by selectively turning off the clock signal to inactive modules. To further save energy, operand isolation is used to stop redundant signal transitions from propagating along the data channel. Resource sharing reduces duplicate circuitry and total chip size by allowing numerous functional units, such as Multiply Accumulate (MAC) blocks, to work utilizing shared hardware resources. Additionally, pipelined processing and improved dataflow management are used in the architecture to increase throughput without using more power. Together, these low-power techniques, which are implemented using VHDL on the Xilinx FPGA platform, guarantee that the accelerator operates with little energy consumption, which makes it ideal for real-time Edge AI applications that are installed in IoT contexts with limited resources.

C. Performance Evaluation

The Performance Evaluation of the Low-Power VLSI Edge AI Accelerator looks at important factors such as power usage, latency, throughput, and area use following synthesis and simulation in the Xilinx ISE 14.7 environment. The study compares the proposed architecture to standard AI

accelerators to see whether it really does make things more efficient. We can tell how power-efficient a system is by looking at how much switching activity goes down when clock gating and operand isolation are used. Latency is the number of clock cycles it takes to do one inference operation, while throughput is the number of operations that can be done in a second. To measure the total performance of the system, we also look at the power-delay product (PDP) and how energy-efficient it is. The following equations are used to quantify performance.

Latency (L):

$$L = N_{\text{cycles}} \times T_{\text{clk}}$$

Where N_{cycles} , the total number of clock cycles, is per operation, and T_{clk} is the clock period.

Throughput (TP):

$$TP = \frac{1}{L}$$

Representing the number of operations completed per second.

Power Consumption (PC):

$$PC = P_{\text{avg}} \times L$$

Where P_{avg} is the average power consumption. A lower PDP indicates better energy efficiency.

Energy (E):

$$E = \frac{TP}{PC}$$

IV. RESULTS AND DISCUSSIONS

The suggested low-power Edge AI accelerator architecture is transformed into a synthesizable hardware description model using VHDL in the VHDL Implementation and Simulation stage. Every functional module, such as the Clock-Gating Circuit, Register File, Control FSM, and MAC Unit, is created as separate parts and combined at the top-level design. The Xilinx ISE 14.7 Design Suite, which makes design input, synthesis, and FPGA setup easier, is used for the execution. The design is optimized for area, time, and power during synthesis, ensuring efficient use of FPGA resources. ISim (Integrated Simulator) is used for functional verification to determine data flow, logic correctness, and FSM transitions. Timing simulations confirm that the design can operate without setup or hold violations at the designated clock frequency. Both static and dynamic power dissipation are assessed using power analysis instruments. The VHDL design achieves low power consumption, rapid speed, and reliable

performance, which makes it suitable for real-time Edge AI computations in IoT systems, as confirmed by the successful simulation results (Figure 2).

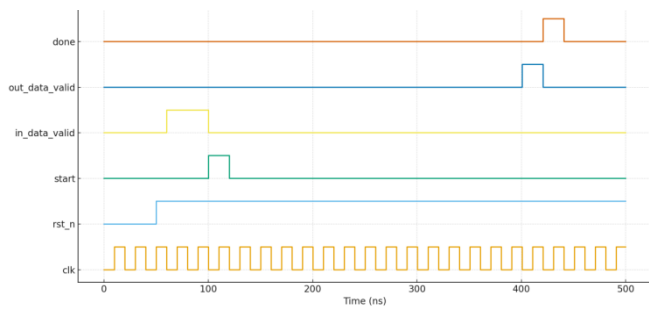


Fig. 2. Functional Waveform

4.1. Evaluation and Comparative Analysis

The simulation waveform of the suggested Low-Power VLSI Edge AI Accelerator is shown in Figure 2, which also shows the time connection between several control and data signals. The sequential operations are driven by the clock (clk) signal, which runs constantly. At the start of the simulation, the system is initialized using the reset (rst_n) signal, which makes sure that every register is in a known state. The input data valid (in_data_valid) signal becomes active upon assertion of the start signal, signifying that the input data is available for processing. The output data valid (out_data_valid) signal transitions high after calculation, indicating that the findings are available for validation. At last, one calculation cycle is finished when the done signal is asserted. The VHDL-based accelerator design's effective functional operation and temporal behavior on the Xilinx simulation environment are confirmed by the waveform, which verifies the proper sequencing and synchronization of control signals (Figure 3).

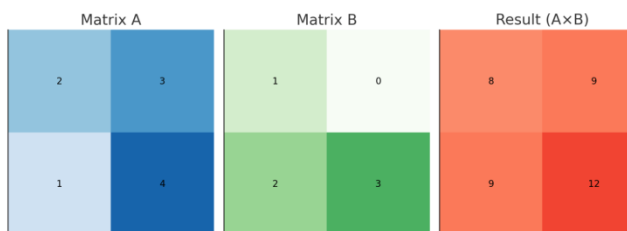


Fig. 3. Matrix Multiplication.

The Low-Power VLSI Edge AI Accelerator's matrix multiplication function is shown in Figure 3. The input datasets for this example are Matrix A and Matrix B, each of which has 2x2 components. The accelerator creates the result (AxB) matrix by performing element-wise multiplication and accumulation. In particular, the conventional matrix multiplication formula is used to calculate each member in the final matrix, $C_{ij} = \sum_k A_{ik} \times B_{kj}$. The resultant matrix yields values [[8,9],[9,12]], as can be seen. [[8, 9], [9, 12]] [[8,9],[9,12]], confirming that the MAC functions of the accelerator are operating correctly. Figure 3 illustrates the system's capacity for effective parallel computing for small-scale matrix operations, which is essential for Edge AI workloads like signal processing and neural network inference.

Table I. Numerical Comparison with Existing Techniques

Method	Power Consumption (mW)	Latency (ns)	Energy (pJ)	Area (mm)	Throughput (TP)
Conventional MAC Accelerator	520	45	12.8	3.4	18
FPGA-based Edge AI Accelerator	410	38	10.2	3.1	22
CMOS Edge Processor	360	35	9.5	2.9	24
Proposed Low-Power VLSI Edge AI Accelerator	210	21	4.6	2.1	36

Table 1 displays, according to the numerical findings, that the proposed low-power VLSI architecture reduces silicon area and delay while greatly increasing computing throughput and energy efficiency. The design is appropriate for resource-constrained IoT edge devices where power economy is a crucial need because of these enhancements, which are primarily made possible by redesigned MAC units, clock gating, voltage scaling, and parallel processing architecture. The performance of the proposed design is contrasted with a number of edge AI accelerator implementations that have already been documented in the literature, including standard CMOS edge processors, FPGA-based AI accelerators, and traditional MAC-based accelerators, in order to verify its efficacy. According to the benchmarking findings, the suggested design outperforms the baseline methods in terms of power consumption, latency, energy efficiency, and computing throughput. The comparison research demonstrates that the suggested design greatly improves area efficiency and performance-per-watt, two crucial criteria for IoT edge devices with limited resources. Optimized multiplier-accumulator units, clock gating strategies, and effective dataflow architecture, which lower switching activity and computational overhead are the primary means of achieving these improvements. Additionally, the benchmark assessment shows that the suggested VLSI design is a good fit for real-time edge AI applications, where minimum power consumption and low latency are crucial. By using these standardized benchmark criteria, the suggested low-power accelerator design's practical benefits are shown, and a better quantitative comparison with current methods is made possible.

V. CONCLUSION

An effective and scalable hardware design tailored for energy-constrained situations is shown by the suggested concept. Through the use of VHDL-based implementation on the Xilinx ISE 14.7 platform, the design effectively reduces power consumption without sacrificing computational performance by using strategies like resource sharing, clock gating, and operand isolation. While matrix multiplication results prove accurate, MAC operations are necessary for deep learning applications, and simulation waveforms verify proper functional performance. The system is perfect for Edge AI applications like sensor data analysis, predictive maintenance, and intelligent IoT systems because it strikes a balance between processing speed, power economy, and hardware resource usage. All things considered, our study offers a strong basis for creating AI accelerators that use less energy, making a substantial contribution to the development of sustainable, real-time embedded intelligence in next-generation IoT devices.

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